

SKILL-2 HIBERNATE CRUD OPERATIONS

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Student.java

```
package com.example;

import jakarta.persistence.Entity;
import jakarta.persistence.Id;

@Entity
public class Student {

    @Id
    private int id;
    private String name;
    private int marks;

    public Student() {}

    public Student(int id, String name, int marks) {
        this.id = id;
        this.name = name;
        this.marks = marks;
    }

    public int getId() { return id; }

    public void setId(int id) { this.id = id; }
```

```
public String getName() { return name; }

public void setName(String name) { this.name = name; }


public int getMarks() { return marks; }

public void setMarks(int marks) { this.marks = marks; }

}
```

App.java

```
package com.example;


import org.hibernate.Session;
import org.hibernate.SessionFactory;
import org.hibernate.cfg.Configuration;


public class App {


    static SessionFactory factory = new Configuration()
        .configure("hibernate.cfg.xml")
        .buildSessionFactory();


    public static void main(String[] args) {


        createStudent();
        readStudent();
        updateStudent();
        deleteStudent();


        factory.close();
    }
}
```

```
// CREATE
```

```
public static void createStudent() {  
    Session session = factory.openSession();  
    session.beginTransaction();  
  
    Student s = new Student(11, "Sai1 Prasad", 95);  
    session.persist(s);  
  
    session.getTransaction().commit();  
    session.close();  
  
    System.out.println("Student Created");  
}
```

```
// READ
```

```
public static void readStudent() {  
    Session session = factory.openSession();  
  
    Student s = session.get(Student.class, 11);  
    if (s != null) {  
        System.out.println("Student Found:");  
        System.out.println(s.getId() + " " + s.getName() + " " + s.getMarks());  
    } else {  
        System.out.println("Student Not Found");  
    }  
  
    session.close();  
}
```

```
}
```

```
// UPDATE
```

```
public static void updateStudent() {
```

```
    Session session = factory.openSession();
```

```
    session.beginTransaction();
```

```
    Student s = session.get(Student.class, 11);
```

```
    if (s != null) {
```

```
        s.setName("Sai Prasad Updated");
```

```
        s.setMarks(98);
```

```
    }
```

```
    session.getTransaction().commit();
```

```
    session.close();
```

```
    System.out.println("Student Updated");
```

```
}
```

```
// DELETE
```

```
public static void deleteStudent() {
```

```
    Session session = factory.openSession();
```

```
    session.beginTransaction();
```

```
    Student s = session.get(Student.class, 11);
```

```
    if (s != null) {
```

```
        session.remove(s);
```

```
    }
```

```
session.getTransaction().commit();

session.close();


System.out.println("Student Deleted");
}
}
```

hibernate.cfg.xml

```
<?xml version="1.0" encoding="utf-8"?>

<!DOCTYPE hibernate-configuration PUBLIC
"-//Hibernate/Hibernate Configuration DTD//EN"
"http://hibernate.sourceforge.net/hibernate-configuration-3.0.dtd">


<hibernate-configuration>

    <session-factory>


        <property name="hibernate.connection.driver_class">
            com.mysql.cj.jdbc.Driver
        </property>

        <property name="hibernate.connection.url">
            jdbc:mysql://localhost:3306/tagdb
        </property>

        <property name="hibernate.connection.username">root</property>
        <property name="hibernate.connection.password">Aaliya@2007</property>


        <property name="hibernate.dialect">
```

```
org.hibernate.dialect.MySQLDialect
```

```
</property>
```

```
<property name="hibernate.hbm2ddl.auto">update</property>
```

```
<property name="hibernate.show_sql">true</property>
```

```
<mapping class="com.example.Student"/>
```

```
</session-factory>
```

```
</hibernate-configuration>
```

Build the Project

```
mvn clean install
```

Run the Application

```
mvn exec:java -Dexec.mainClass="com.example.App"
```

OUTPUT:

```
Hibernate: insert into Student (marks,name,id) values (?,?,?)
Student Created
Hibernate: select s1_0.id,s1_0.marks,s1_0.name from Student s1_0 where s1_0.id=?
Student Found:
11 Sai1 Prasad 95
Hibernate: select s1_0.id,s1_0.marks,s1_0.name from Student s1_0 where s1_0.id=?
Hibernate: update Student set marks=?,name=? where id=?
Student Updated
Hibernate: select s1_0.id,s1_0.marks,s1_0.name from Student s1_0 where s1_0.id=?
Hibernate: delete from Student where id=?
Student Deleted
```